Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

J Stud Alcohol Drugs. Author manuscript; available in PMC 2026 April 20.

E-Cigarette Use Trajectories and Respiratory Symptoms in Adolescents and Young Adults: Sustained Use Linked to Greater Symptom Frequency

Abigail Shores, BSPH[1], **Hannah Appleseth, MA**[2], **Ruochen Zhao, PhD**[3], **Megan E. Roberts, PhD**[1,4], **Bo Lu, PhD**[1], **Jill M. Singer, PhD, MPH**[1], **Dylan Wagner, PhD**[5], **Elizabeth G. Klein, PhD, MPH**[1,4], **Loren E. Wold, PhD**[6], **Marielle C. Brinkman, BS**[1,4], **Theodore L. Wagener, PhD**[4,7], **Alayna P. Tackett, PhD**[4,7]

[1]College of Public Health, The Ohio State University, Columbus, Ohio, USA

[2]Department of Psychology, Oklahoma State University, Stillwater, Oklahoma, USA

[3]Center for Biologics Evaluation and Research, U.S. Food and Drug Administration, Silver Spring, Maryland, USA

[4]Center for Tobacco Research, The Ohio State University Comprehensive Cancer Center, Columbus, Ohio, USA

[5]Department of Psychology, The Ohio State University, Columbus, Ohio, USA

[6]Department of Surgery, The Ohio State University, Columbus, Ohio, USA

[7]Department of Internal Medicine, The Ohio State University Medical Center, Columbus, Ohio, USA

Abstract

Objective: This study cross-sequentially examined three measures of self-reported respiratory symptoms and nicotine e-cigarettes (EC) use adjusting for past 30-day use of cannabis/tobacco, COVID-19, and lifetime asthma at baseline (T1) and 12-months (T2).

Method: Data came from the Emerging Adulthood Health Project. AYAs aged 15–24 (N= 751) self-reported demographics, nicotine/cannabis use, and respiratory symptoms. EC use at T1-T2 was categorized into a) never use, b) non-escalating, c) increasing, or d) sustained use. Binary and linear logistic regressions examined outcomes at T2.

Results: At T1, participants (M age= 20.56 years [SD= 2.09 years]) were mostly female (n= 423 [56.6%]), White (n= 563 [75.3%]), and college students (n= 510 [68.0%]). In unadjusted analyses, EC sustained use had a significantly higher number of summed symptoms (binary; OR= 1.47, 95% CI [1.04, 2.07], p= .03) and more days with symptoms (OR= 1.64, 95% CI [1.16,

Correspondence: Alayna P. Tackett, PhD, The Center for Tobacco Research, Department of Internal Medicine, The Ohio State University Wexner Medical School, 3650 Olentangy River RD Suite 420, Columbus OH 43214. Alayna.Tackett@osumc.edu.

Author Agreement

We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. We have no conflicts of interest to disclose. All authors participated in the development of this manuscript, have agreed to the order of authorship on the byline, and to submission of the manuscript in this form.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

2.32], p= .006) compared to never EC use. In adjusted analyses, sustained EC use had higher American Thoracic Society Questionnaire scores (b=1.48, SE= 0.54, p=.0006) than never EC use, but the number of summed symptoms and days with symptoms became non-significant. Among participants experiencing any day with symptoms (N= 269), sustained (b= 2.05, SE= 0.95, p=.03) and non-escalating use (b= 2.73, SE= 1.17, p= .02) was associated with more days with symptoms than never EC use in adjusted analyses.

Conclusions: EC use behavior impacts self-reported frequency of respiratory symptoms among AYAs, with those who continue to use reporting more frequent symptoms, after adjusting for tobacco/cannabis use, COVID-19, and asthma.

Despite declines in overall past 30-day e-cigarette use among adolescents and young adults (AYAs) in 2023 (12.6%) and 2024 (10.1%), e-cigarettes remain the most common nicotine product used and tried by high school and middle school students in the United States (Jamal, 2024). While prevalence of nicotine vaping among U.S. youth declined between 2020 and 2024, those who continue to use show higher rates of daily use and reduced cessation success, consistent with increasing nicotine dependence (Masonbrink et al., 2025). Preliminary evidence suggests e-cigarette use can increase cardiovascular risk, contain oxidants, aldehydes, and harmful metals, which are known respiratory irritants, (Chaiton et al., 2023; Lee et al., 2017; Lerner et al., 2015) and is associated with respiratory symptoms and illnesses (Bhatta & Glantz, 2020; Chaiton et al., 2023; Rose et al., 2023; Wang et al., 2018). Adolescence and young adulthood are critical periods for nicotine use initiation (Barrington-Trimis et al., 2020). Cross-sectional studies found AYA e-cigarette use was associated with self-reported wheeze, shortness of breath, dry cough and other indicators of pulmonary distress after controlling for asthma diagnosis and other tobacco product use (Cherian et al., 2021; Cho & Paik, 2016; Gotts et al., 2019; D. Li et al., 2020; McConnell et al., 2017; W. Xie et al., 2022).

Prospective studies suggest AYA e-cigarette use is associated with increased risk of bronchitic symptoms, shortness of breath, and wheezing, while controlling for covariates of cannabis and other tobacco product use (Chaffee et al., 2021; Tackett et al., 2023) but some longitudinal studies have found mixed results (Chaffee et al., 2021; Stevens et al., 2022; Tackett et al., 2023). Analyses of Waves 1–5 (2013–2019) of the PATH study found that exclusive e-cigarette use (incident rate ratio=1.49, 95% CI=1.06, 2.08) was associated with increased diagnosed bronchitis, pneumonia, or chronic cough than never/noncurrent use (Mukerjee et al., 2024). However, analyses of Waves 3 and 4 from another longitudinal PATH study found that youth who reported e-cigarette use at Wave 3 did not have increased odds of having functionally important (2 or more) respiratory symptoms at the follow-up time point (Wave 4) (Stevens et al., 2022). However, this study did not account for changes in e-cigarette use over time.

Current Study

This cross-sequential study examined how changes in e-cigarette use over a one-year period are associated with self-reported respiratory symptoms among a sample of mostly college enrolled AYAs from baseline (2021; T1) to a 12-month follow-up (2022; T2). To capture the dynamic nature of e-cigarette use among AYAs, we developed four unique categories: never

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

use, non-escalating, increasing, and sustained use. These categories are based solely on change in e-cigarette behavior from T1 to T2 and do not account for quantity of e-cigarette use or other tobacco products. This categorization approach aims to provide a more nuanced understanding of AYA-specific behaviors compared to other approaches which rely on lifetime (ever use) or 1+ days used in past 30-day use (current use) (Boyle et al., 2021; Dutra & Glantz, 2017; Gagné et al., 2021). Similar data suggest that AYA use of other tobacco products and e-cigarettes is complex and that frequency or intensity of use may change over time (Soule et al., 2023; Villanti et al., 2017; C. Xie et al., 2024).

Three measures of respiratory outcomes were utilized to comprehensively examine respiratory symptoms. First, total summed respiratory symptoms (i.e., “summed symptoms”) were examined to associate use with a variety of respiratory outcomes (e.g., cough, chest tightness, chest pain, shortness of breath). Second, we utilized the total number of days in the past 30-days in which each symptom occurred (i.e., “number of days with symptoms in the past 30-days;” 0–30 days). Finally, the American Thoracic Society Questionnaire [ATSQ] total score determined the occurrence of a variety of respiratory symptoms (e.g., phlegm or mucous when you cough, pain or tightness in the chest) over the past week (Cassidy et al., 2015). We hypothesized a) summed symptoms, b) number of days with symptoms in the past 30-days, and c) reoccurrence of ATSQ symptoms would be higher among AYAs who sustained e-cigarette use or increased use between T1 and T2 compared to AYAs who reported never use of e-cigarettes.

Method

Participants and Procedures

Data were obtained from Emerging Adulthood Health Project, an American Heart Association funded longitudinal survey project. This cohort study was established to assess e-cigarette and tobacco use among AYAs in Ohio. Participants were recruited from January 2021 to September 2021 using online (e.g., social media, university website) and print (e.g., flyers) advertisements. Participants aged 15–24 and currently residing in Ohio (verified via county and zip code) were eligible for the study. Quota sampling ensured adequate representations by gender, race, and obtained an approximately 50/50 ratio of never tobacco use and current e-cigarette use.

Online consent was obtained for adults over 18. Verbal assent and parental permission were obtained from participants aged 15–17 years. Following informed consent, participants were asked to complete a baseline questionnaire followed by online follow up surveys at 4-months, 8-months, and 12-months in 2022. Data from baseline (T1) and 12-month follow-up (T2) were used to categorize use groups. Covariates were defined at baseline, and respiratory symptoms at the 12-month point. Participants were provided with $25 for each assessment via gift card. This study was approved by the University’s Institutional Review Board.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Measures

Sample Characteristics: Participants self-reported sociodemographic information including date of birth, gender identity (recoded as male, female, or another gender response), and race and ethnicity (recoded as Hispanic, or non-Hispanic; White, Black, Native American, Asian, Pacific Islander and multi-racial).

Past 30-Day Cannabis Use: Participants were asked, "How long has it been since you last used marijuana?" (today, sometime in the last week, sometime in the last month, sometime in the last 4 months, sometime in the last 6 months, more than 6 months ago, or more than a year ago). Respondents who indicated cannabis use a) today, b) sometime in the last week, or c) sometime in the last month) were categorized as having used cannabis in the Past 30-Days. A follow up question asked, "In what ways do you use marijuana?" (smoke it (e.g., joint), vape it, eat it (edibles), drink it (juice), dab it (wax) and other). All options of "smoke", "vape", and "wax" were combined to create the "Inhalable Cannabis" variable while "edibles" and "juice" were combined to create an "Edible Cannabis" variable. Cannabis modalities were not mutually exclusive.

Past 30-Day Cigarette and Other Tobacco Product Use: All participants indicated which tobacco products, if any, they had used in the past 30-days (cigarettes, cigars, cigarillos, smokeless tobacco, hookah, pipe tobacco). Participants who responded with a) every day, or almost every day, b) at least once a week, or c) at least once a month for any tobacco products were categorized as having used Tobacco in the Past 30-Days. This included both combustible and non-combustible tobacco products, excluding e-cigarettes.

E-cigarette Use: Participants were also asked "During the past 30 days, on how many days did you use an e-cigarette/vaping device?". Any non-zero responses were categorized as Past-30 Day e-cigarette use. Respondents completed the same questionnaire at T1 and T2, except at T1 they were asked about past 3-month e-cigarette use and at T2 they reported on past 4-month e-cigarette use, as 4 months had elapsed from the previous survey (8-month follow up).

Cross-Sequential E-cigarette Use Categorization (Figure 1). All participants who fully completed both T1 and T2 surveys were sorted into an e-cigarette use category. E-cigarette use was categorized cross-sequentially into a) sustained use (past 30-day use at T1 and T2), b) non-escalating (past 30-day at T1 to 4 month or ever use at T2; past 4-month at T1 to ever use at T2; never at T1 to ever use at T2, ever at both T1 and T2), c) increasing (ever or never at T1 to past 4- month use or past 30-day use at T2, past 4-month at T1 to past 30-day use at T2) or d) never use (never used e-cigarettes at T1 or T2).

Summed Respiratory Symptoms: Respiratory symptoms were assessed using items from the Population Assessment of Tobacco and Health (PATH) (Hyland et al., 2017) questionnaire as well as commonly used items from the published literature (Cherian et al., 2021; D. Li et al., 2020; Stevens et al., 2022; W. Xie et al., 2022). Items included six symptoms: cough, chest tightness, chest pain, shortness of breath, sore or dry mouth and throat, and breathing difficulties. Participants self-reported (Yes/No) if they experienced

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

each symptom in the past three months at both T1 and T2. The summed respiratory symptoms variable (i.e., “summed symptoms”) is total number of these symptoms (0 symptoms to 6 symptoms; linear regression model) at T2 and was dichotomized into a binary variable whereas participants either had symptoms (1) or did not have any symptoms (0) at T2 for the binary logistic regression model.

Number of Days with Respiratory Symptoms in the Past 30-Days: For each of the respiratory symptoms (listed above), participants were asked to report the number of days in the past 30-days each symptom occurred (i.e., “number of days with symptoms”). The number of days were totaled and averaged to provide a mean of the number of days in the past 30-days on which they experienced at least one respiratory symptom at T2 for the linear regression model.

For the binary logistic model, past 30-day symptoms at T2 were then dichotomized into a binary variable where participants either experienced any day with respiratory symptoms (1) or experienced no days with respiratory symptoms (0) at T2 for the logistic regression model.

American Thoracic Society Questionnaire (ATSQ): The ATSQ (Cassidy et al., 2015) is an 8-item questionnaire used to assess the occurrence of respiratory symptoms over the past week (e.g., cough first thing in the morning) with response options categorized as (1) never; (2) less than once per week; (3) 1–2 times per week; (4) several times per week; and (5) every day (Cassidy et al., 2023). The ATSQ provided a total score which ranges from 8 to 40, with higher scores indicating more frequent and recurrent respiratory symptoms. Total scores at T2 were examined in the linear regression model. For the binary logistic regression model, scores were dichotomized into a binary variable which categorized participants as either having (1) any ATSQ symptoms throughout the week or (0) not having any ATSQ symptoms throughout the week at T2.

Lifetime Asthma Diagnosis and COVID-19 History: Participants self-reported medical professional diagnosed asthma (Yes/No). “Yes” were categorized as individuals with a lifetime asthma diagnosis. Participants were also asked if they have or have ever had COVID-19 (Yes/No). “Yes” at T1 were categorized as participants with a self-reported COVID-19 history.

Statistical Analysis

Descriptive statistics were used to assess the distribution of baseline covariates (past 30-day other tobacco use, past 30-day cannabis use, COVID-19, and lifetime asthma diagnosis) and outcome variables across each e-cigarette use category (never, non-escalating, increasing, and sustained use). Two-part regression models examined the associations between e-cigarette use categories and respiratory symptoms. The first component used a logistic regression model to estimate the odds of reporting any respiratory symptoms (0=*No Symptoms*, 1=*Symptoms*). The second component used a zero-inflated linear regression model to quantify the frequency of respiratory symptoms among participants who experienced non-zero values for both number of days in the past 30- days and average ATSQ

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

score. This two-part modeling approach is commonly used in health outcomes research to address zero-inflated symptom distributions (Farewell et al., 2017; Yang et al., 2016).

Previous studies have shown that dual-use of other tobacco products, (Cordova et al., 2022) cannabis, (Correa et al., 2020; Winhusen et al., 2019) and a lifetime asthma diagnosis (Benidis et al., 2023; Cho & Paik, 2016; X. Li et al., 2022) can impact self-reported respiratory symptoms and were used as covariates in the present study. Individuals with a lifetime asthma diagnosis were excluded from ATSQ analyses as suggested by prior research (Cassidy et al., 2023). Sociodemographic covariates were evaluated in inferential models but were non-significant and excluded from final analyses. All analyses were performed using R Statistical Software (v4.2.3; R Core Team 2023), and all statistical tests were two-sided with a significance level set at $\alpha = 0.05$.

Results

Participant Characteristics (Table 1)

A total of 1068 AYAs were recruited, however, 317 participants (29.7%) did not complete the 12-month follow-up or did not answer all required questions and were excluded from the analytic sample. $N = 751$ participants with complete information on e-cigarette use and respiratory outcomes were included in the analytic sample. Participants were majority female ($n = 423$, 56.6%), Non-Hispanic White ($n = 563$, 75.3%), in college ($n = 510$, 68.0%) and had an average age of 20.56 years ($SD = 2.09$). Across the sample, (19.7%) ($n = 148$) of participants reported having or had COVID-19 at T1 but COVID-19 status was not significantly associated with any of the outcome measures.

Substance Use and Descriptives

Across the analytic sample, 28.1% ($n = 211$) reported sustained e-cigarette use, 15.9% ($n = 120$) reported non-escalating use, 6.5% ($n = 49$) reported increased use, and 49.4% ($n = 371$) never used an e-cigarette at T1 and T2. Cannabis use (T1: $n = 165$ [22.2%]; T2: $n = 180$ [24.0%]) and past 30-day cigarette and other tobacco use (T1: $n = 69$ [9.2%]; T2: $n = 58$ [7.7%]) were similar at both time-points. The majority of the past 30-day cannabis use participants were inhaling cannabis (T1: $n = 160$ [97.0%]; T2: $n = 154$ [85.6%]).

Table 1 provides a summary of cigarette and other tobacco product use and/or cannabis stratified among the e-cig use categories. Frequency varied across assessment timepoints, whereas participants in the sustained e-cigarette use category reported the highest frequency of cigarette and other tobacco use (T1: $n = 51$ [24.2%]; T2: $n = 49$ [23.2%]) and also reported cannabis use more often at T2 ($n = 115$ [54.5%]) than T1 ($n = 109$ [51.7%]). While cigarette and other tobacco use was relatively low in the non-escalating e-cigarette use category (T1: $n = 16$ [13.3%]; T2: $n = 4$ [3.3%]), cannabis use at T1($n = 55$ [45.8%]) was relatively high but decreased at T2 ($n = 46$ [38.3%]). While participants in the increasing e-cigarette use category reported increases in their e-cigarette use frequency, their cigarette and other tobacco use (T1: $n = 2$ [4.1%]; T2: $n = 5$ [10.2%]) and cannabis use (T1: $n = 1$ [2.0%]; T2: $n = 5$ [10.2%]) remained relatively low.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 1 also provides the descriptive statistics for the summed symptom score, number of days with symptoms, and ATSQ score, stratified by e-cigarette use category. Across the analytic sample, participants reported an average of 0.89 (*SD*= 1.32) summed symptoms (scale of 0–6), an average of 2.2 days (*SD*= 4.3 days) in the past 30-days with respiratory symptoms (scale of 0–30), and an ATSQ score of 12.0 (*SD*= 4.04), (scale of 8–40) at T1. Results were similar at T2 with an average of 0.79 (*SD*= 1.26) summed symptoms, an average of 2.1 days (*SD*= 4.4 days) in the past 30-days with respiratory symptoms, and an ATSQ score of 11.6 (*SD*= 4.16).

Table 2 shows the specific respiratory symptom item frequencies from the summed symptom and ATSQ measures, stratified by e-cigarette use category. Across the analytic sample, the most endorsed symptom within the summed symptom questionnaire was cough (T1: *n*= 203 [27%]; T2: *n*= 193 [25.7%]). For the ATSQ, the most reported symptom was shortness of breath during exercise or walking upstairs (T1: *n*= 254 [33.8%]; T2: *n*= 208 [27.7%]).

Cross-Sequential E-Cigarette Use Categories and Respiratory Outcome Analyses

In initial analyses, all three respiratory outcomes were highly skewed and did not fit a normal distribution, requiring dichotomization. Linear regressions were used to assess symptom frequency among participants with non-zero number of days with symptoms in the past 30-days (*N*= 269) and ATSQ scores (*N*= 420).

Summed Symptoms (Tables 3 and 4)

In unadjusted logistic regression analyses, participants who sustained e-cigarette use had significantly higher odds of experiencing any respiratory symptoms at T2 compared to participants who never used (logistic regression (Table 3); odds ratio [OR]= 1.47, 95% CI [1.04, 2.07], *p*= .03). In adjusted logistic regression models this effect was no longer significant (OR= 1.18, 95% CI [0.76, 1.82], *p*= .46). No significant differences were observed for never-use, non-escalating, or increasing use categories in unadjusted or adjusted analyses.

Number of Days with Symptoms in the Past 30-Days (Tables 3 and 4)

In unadjusted logistic regression analyses, participants who sustained e-cigarette use had significantly greater odds (OR= 1.64, 95% CI [1.16, 2.32], *p*= .006) of experiencing a day with respiratory symptoms in the past 30-days compared to participants who never used. In adjusted logistic regression analyses, these results were no longer significant (aOR= 1.22, 95% CI [0.92, 1.97], *p*= 0.37). No significant differences were observed across the other use categories (Table 3).

In adjusted linear regression models (Table 4), among participants who reported any day with respiratory symptoms in the past 30-days (*N*= 269), both those who reported sustained use of e-cigarettes (*b*= 2.05 days, *SE*= 0.95, *p*= .03) and non-escalating use of e-cigarettes (*b*= 2.73 days, SE= 1.17, *p*= .02) reported a significantly higher average of days with symptoms compared to participants who never used e-cigarettes.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

American Thoracic Society Questionnaire (Tables 3 and 4)

In logistic regression models (Table 3), no significant associations were observed for any e-cigarette use category in both unadjusted and adjusted models (p's= .24 – 61). To explore differences in ATSQ scores among participants with at least 1 ATSQ symptom (i.e., minimum score of 9) (N= 420) an adjusted linear regression model (Table 4) was used to examine differences by e-cigarette use category controlling for covariates. Similar to the other respiratory outcomes, participants in the sustained use category had significantly higher ATSQ scores (b= 1.48, SE= 0.54, p= .0006) compared to participants who never used e-cigarettes.

Discussion

Young nicotine users often have dynamic use patterns and may decline or escalate in their use during different periods of their lives (Bandi et al., 2021; Niaura et al., 2020; Stanton et al., 2020; Villanti et al., 2017). Previous studies examining e-cigarette use among AYAs classified use with categories like "ever use" (≥1 puff) and "current use" (at least 1 day within the last 30 days) (Dutra & Glantz, 2017) or with lifetime behavior groupings like "daily," "non-daily," "former," or "never" (Boyle et al., 2021; Gagné et al., 2021). In this study, we categorized use status by how use changed cross-sequentially between T1 and T2. We also examined three outcomes of respiratory symptoms to determine if different approaches better captured self-reported symptoms in this demographic. In general, participants who sustained e-cigarette use reported a greater number of symptom days in the past 30-days and a greater number of respiratory symptoms (summed symptoms) in unadjusted analyses compared to AYAs who never used e-cigarettes. However, after controlling for covariates that likely impact respiratory functioning (i.e., lifetime asthma diagnosis, COVID-19, use of cigarettes and other tobacco products and/or cannabis), findings were no longer significant in the binary logistic models. Adjusted linear regressions among participants experiencing respiratory symptoms found that participants who reported sustained e-cigarette use experienced more frequent respiratory symptoms (higher ATSQ scores) and a greater number of days with respiratory symptoms in the past 30-days compared to participants who never used e-cigarettes. Unexpectedly, participants in the non-escalating use category reported a greater frequency (2.73 days) of symptoms in the past 30-days compared to participants who never used. Our findings suggest that frequency of respiratory symptoms related to e-cigarette use likely varies depending on continued use of e-cigarettes and is impacted by factors such as COVID-19, preexisting respiratory conditions (asthma) and use of cigarette and other tobacco and/or cannabis products.

While these results were expected for the sustained use category and are consistent with the extant literature to date, (Chaiton et al., 2023) the non-escalating use participants experiencing more symptoms days is an interesting finding. It could be that AYAs in the non-escalating category were already experiencing respiratory symptoms and, for that reason, did not escalate their e-cigarette use. Similarly, participants in this category also showed declining use across all products (45.8% to 38.3% cannabis; 13.3% to 3.3% tobacco). This is notable given the high frequency of tobacco and cannabis use in our sample. Participants who sustained e-cigarette use exhibited possible co-use behaviors, with

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

cannabis use remaining consistent (51.7% to 54.5%) and cigarettes and other tobacco use showing minimal change (24.2% to 23.2%) from T1 to T2. Participants who increased use exhibited rising trends (2.0% to 10.2% cannabis; 4.1% to 10.2% tobacco), aligning with national data showing increased cannabis and nicotine vaping among young adults (Liu et al., 2023).

Data are emerging about the respiratory impact of dual-using e-cigarettes and cannabis, with some cross-sectional data suggesting that vaping both nicotine and cannabis is associated with respiratory symptoms of wheeze and bronchitic symptoms (Braymiller et al., 2020). Previous studies suggest that AYAs who use tobacco, nicotine, and co-use of other products like cannabis often switch between products, thus leading to more unpredictable use patterns (National Institutes of Health [NIH]., 2024; Barrington-Trimis et al., 2018; Niaura et al., 2020; Reboussin et al., 2021; Weinberger et al., 2020). For example, data from the 2025 Monitoring the Future found that prevalence of cannabis use among high school seniors (25.7%) and youth in 8th grade (7.6%) remained substantial despite a decrease from pre-pandemic levels in 2020, while e-cigarette use decreased in all grades for lifetime, past 12-month, and past 30-day use (Miech, R. A., et al., 2026). A 2023 study with a similar demographic of AYAs in New England found that past 30-day vaping of both nicotine and cannabis (7.3%) was more prevalent than past 30-day vaping of either nicotine (5.5%) or cannabis (5.6%) alone (Liu et al., 2023). The complexity of multiple substance use among AYAs underscores the need for intensive approaches to studying e-cigarette use and its health impact in young populations, while recognizing that possible co-use behaviors are common, change rapidly, and may be increasing in prevalence.

Limitations and Future Directions

While this cross-sequential study of e-cigarette use and respiratory symptoms among AYAs has several strengths, including a large sample size, use of three well-validated measurement questionnaires to capture respiratory symptoms, and a more descriptive categorization of e-cigarette use over a 1-year period, it also has limitations. First, as the majority of our participants were enrolled in college or advanced degree programs, the sample may not fully capture the experiences of adolescents and young adults (AYA) who are not pursuing higher education, or who live outside of Ohio. Second, while this study does assess for cigarette and other tobacco product and cannabis use, it does not directly assess the effects of verified co-use patterns as documented in current literature (Nguyen et al., 2024). Third, while previous work has utilized similar methodologies to examine respiratory outcomes, (Chaffee et al., 2021; Stevens et al., 2022; Tackett et al., 2023) self-reported symptoms may result in data with issues of recall bias and social desirability (Althubaiti, 2016). Lastly, it should be noted that questionnaires were self-administered online during the COVID-19 pandemic, possibly leading to higher rates of attrition (29.7%) and potential long COVID impacts on overall respiratory health. Building on these findings, future research can incorporate biochemical verification of use and objective measures of respiratory health.

Conclusion

This study highlights the dynamic nature of e-cigarette use among adolescents and young adults (AYAs) over a one-year period. While the likelihood of experiencing any respiratory

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

symptoms did not significantly differ by e-cigarette use category in adjusted models, symptom frequency did. Participants who engaged in past-30-day e-cigarette use at both timepoints ("sustained use") reported more symptomatic days and higher ATSQ scores than participants who never used e-cigarettes, while participants who had non-escalating use also experienced more symptomatic days compared to participants who never used. Our findings add to the growing literature suggesting that AYAs employ dynamic patterns of e-cigarette use and often engage in multiple product use (Barrington-Trimis et al., 2018; Niaura et al., 2020; Reboussin et al., 2021). These findings emphasize the need for future research as AYA nicotine behavior may require individualized approaches to prevent escalation of multi-product use.

Acknowledgement

This article reflects the views of the authors and should not be constructed to represent NIH, AHA, or FDA's views or policies. The funders had no role in the design and conduct of the study; collection, management, analysis and interpretation of the data; preparation, review or approval of the manuscript; or decision to submit the manuscript for publication.

Funding

This study was supported by the American Heart Association under award number 20YVNR35490079 (Megan E. Roberts, Alayna P. Tackett), K01HL148907 (Alayna P. Tackett).

Data availability

Data supporting the conclusions of this manuscript will be made available upon request.

References

Althubaiti A (2016). Information bias in health research: Definition, pitfalls, and adjustment methods. Journal of Multidisciplinary Healthcare, 9, 211–217. 10.2147/JMDH.S104807 [PubMed: 27217764]

Bandi P, Cahn Z, Goding Sauer A, Douglas CE, Drope J, Jemal A, & Fedewa SA (2021). Trends in E-Cigarette Use by Age Group and Combustible Cigarette Smoking Histories, U.S. Adults, 2014–2018. American Journal of Preventive Medicine, 60(2), 151–158. 10.1016/j.amepre.2020.07.026 [PubMed: 33032869]

Barrington-Trimis JL, Braymiller JL, Unger JB, McConnell R, Stokes A, Leventhal AM, Sargent JD, Samet JM, & Goodwin RD (2020). Trends in the Age of Cigarette Smoking Initiation Among Young Adults in the US From 2002 to 2018. JAMA Network Open, 3(10), e2019022. 10.1001/jamanetworkopen.2020.19022 [PubMed: 33021650]

Barrington-Trimis JL, Kong G, Leventhal AM, Liu F, Mayer M, Cruz TB, Krishnan-Sarin S, & McConnell R (2018). E-cigarette Use and Subsequent Smoking Frequency Among Adolescents. Pediatrics, 142(6), e20180486. 10.1542/peds.2018-0486 [PubMed: 30397165]

Benidis KD, Tzortzaki E, Georgiou A, Zachariadou T, Adamidi T, Zannetos S, Bakakos P, Koulouris NG, & Rovina N (2023). Prevalence and Characteristics of Self-Reported Adult Asthma in Cyprus: A Population-Based Observational Study. Journal of Asthma and Allergy, 16, 215–226. 10.2147/JAA.S397784 [PubMed: 36874225]

Bhatta DN, & Glantz SA (2020). Association of E-Cigarette Use With Respiratory Disease Among Adults: A Longitudinal Analysis. American Journal of Preventive Medicine, 58(2), 182–190. 10.1016/j.amepre.2019.07.028 [PubMed: 31859175]

Boyle RG, Richter S, & St Claire AW (2021). Defining adult e-cigarette prevalence: Comparing a categorical definition with days of use. Tobacco Control, 30(5), 530–533. 10.1136/tobaccocontrol-2020-055641 [PubMed: 32675251]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Braymiller JL, Barrington-Trimis JL, Leventhal AM, Islam T, Kechter A, Krueger EA, Cho J, Lanza I, Unger JB, & McConnell R (2020). Assessment of Nicotine and Cannabis Vaping and Respiratory Symptoms in Young Adults. JAMA Network Open, 3(12), Article 12. 10.1001/jamanetworkopen.2020.30189

Cassidy RN, Roberts ME, & Colby SM (2015). Validation of a Respiratory Symptom Questionnaire in Adolescent Smokers. Tobacco Regulatory Science, 1(2), 121–128. 10.18001/TRS.1.2.2 [PubMed: 26236764]

Cassidy RN, Tidey JW, Jackson KM, Cioe PA, Murphy SE, Krishnan-Sarin S, Hatsukami D, & Colby SM (2023). The Impact of Reducing Nicotine Content on Adolescent Cigarette Smoking and Nicotine Exposure: Results From a Randomized Controlled Trial. Nicotine & Tobacco Research, 25(5), 918–927. 10.1093/ntr/ntac279 [PubMed: 36482794]

Chaffee BW, Barrington-Trimis J, Liu F, Wu R, McConnell R, Krishnan-Sarin S, Leventhal AM, & Kong G (2021). E-cigarette use and adverse respiratory symptoms among adolescents and Young adults in the United States. Preventive Medicine, 153, 106766. 10.1016/j.ypmed.2021.106766 [PubMed: 34418439]

Chaiton M, Pienkowski M, Musani I, Bondy SJ, Cohen JE, Dubray J, Eissenberg T, Kaufman P, Stanbrook M, & Schwartz R (2023a). Smoking, e-cigarettes and the effect on respiratory symptoms among a population sample of youth: Retrospective cohort study. Tobacco Induced Diseases, 21, 08. 10.18332/tid/156839 [PubMed: 36721859]

Chaiton M, Pienkowski M, Musani I, Bondy SJ, Cohen JE, Dubray J, Eissenberg T, Kaufman P, Stanbrook M, & Schwartz R (2023b). Smoking, e-cigarettes and the effect on respiratory symptoms among a population sample of youth: Retrospective cohort study. Tobacco Induced Diseases, 21, 08. 10.18332/tid/156839 [PubMed: 36721859]

Cherian C, Buta E, Simon P, Gueorguieva R, & Krishnan-Sarin S (2021). Association of Vaping and Respiratory Health among Youth in the Population Assessment of Tobacco and Health (PATH) Study Wave 3. International Journal of Environmental Research and Public Health, 18(15), Article 15. 10.3390/ijerph18158208

Cho JH, & Paik SY (2016). Association between Electronic Cigarette Use and Asthma among High School Students in South Korea. PLOS ONE, 11(3), e0151022. 10.1371/journal.pone.0151022 [PubMed: 26942764]

Cordova J, Pfeiffer RM, Choi K, Mayne RG, Baker L, Bachand J, Constantine K, Altekruse S, & Reyes-Guzman C (2022). Tobacco use profiles by respiratory disorder status for adults in the wave 1-wave 4 population assessment of tobacco and health (PATH) study. Preventive Medicine Reports, 30, 102016. 10.1016/j.pmedr.2022.102016 [PubMed: 36325251]

Correa JB, Myers MG, Tully LK, & Doran N (2020). Co-occurring Use of Cannabis and Tobacco and the Presence of Acute Respiratory Symptoms among Young Adult Light and Intermittent Smokers. Substance Use & Misuse, 55(13), 2129–2137. 10.1080/10826084.2020.1793366 [PubMed: 32696699]

Dutra LM, & Glantz SA (2017). E-cigarettes and National Adolescent Cigarette Use: 2004–2014. Pediatrics, 139(2), e20162450. 10.1542/peds.2016-2450 [PubMed: 28115540]

Farewell VT, Long DL, Tom BDM, Yiu S, & Su L (2017). Two-Part and Related Regression Models for Longitudinal Data. Annual Review of Statistics and Its Application, 4, 283–315. 10.1146/annurev-statistics-060116-054131

Gagné T, Lee JGL, & O'Loughlin J (2021). Cigarette smoking and E-cigarette use among young adults in the United States: Findings from the 2016–18 behavioral risk factor surveillance system. Addictive Behaviors, 113, 106675. 10.1016/j.addbeh.2020.106675 [PubMed: 33038680]

Gotts JE, Jordt S-E, McConnell R, & Tarran R (2019). What are the respiratory effects of e-cigarettes? BMJ, 366, l5275. 10.1136/bmj.l5275 [PubMed: 31570493]

Hyland A, Ambrose BK, Conway KP, Borek N, Lambert E, Carusi C, Taylor K, Crosse S, Fong GT, Cummings KM, Abrams D, Pierce JP, Sargent J, Messer K, Bansal-Travers M, Niaura R, Vallone D, Hammond D, Hilmi N, … Compton WM (2017). Design and methods of the Population Assessment of Tobacco and Health (PATH) Study. Tobacco Control, 26(4), 371–378. 10.1136/tobaccocontrol-2016-052934 [PubMed: 27507901]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Jamal A (2024). Tobacco Product Use Among Middle and High School Students—National Youth Tobacco Survey, United States, 2024. MMWR. Morbidity and Mortality Weekly Report, 73. 10.15585/mmwr.mm7341a2

Lee M-S, LeBouf RF, Son Y-S, Koutrakis P, & Christiani DC (2017). Nicotine, aerosol particles, carbonyls and volatile organic compounds in tobacco- and menthol-flavored e-cigarettes. Environmental Health, 16(1), 42. 10.1186/s12940-017-0249-x [PubMed: 28449666]

Lerner CA, Sundar IK, Watson RM, Elder A, Jones R, Done D, Kurtzman R, Ossip DJ, Robinson R, McIntosh S, & Rahman I (2015). Environmental health hazards of e-cigarettes and their components: Oxidants and copper in e-cigarette aerosols. Environmental Pollution, 198, 100–107. 10.1016/j.envpol.2014.12.033 [PubMed: 25577651]

Li D, Sundar IK, McIntosh S, Ossip DJ, Goniewicz ML, O'Connor RJ, & Rahman I (2020). Association of smoking and electronic cigarette use with wheezing and related respiratory symptoms in adults: Cross-sectional results from the Population Assessment of Tobacco and Health (PATH) study, wave 2. Tobacco Control, 29(2), 140–147. 10.1136/tobaccocontrol-2018-054694 [PubMed: 30760629]

Li X, Zhang Y, Zhang R, Chen F, Shao L, & Zhang L (2022). Association Between E-Cigarettes and Asthma in Adolescents: A Systematic Review and Meta-Analysis. American Journal of Preventive Medicine, 62(6), 953–960. 10.1016/j.amepre.2022.01.015 [PubMed: 35337694]

Liu J, Tan ASL, Winickoff JP, & Rees VW (2023). Correlates of adolescent sole-, dual- and poly-use of cannabis, vaped nicotine, and combusted tobacco. Addictive Behaviors, 146, 107804. 10.1016/j.addbeh.2023.107804 [PubMed: 37478525]

Masonbrink AR, Bae D, Cho J, Miech RA, Dai HD, Harlow AF, Sussman S, Han D-H, Sanchez LM, Adjei A, Meza LR, Li M, & Leventhal AM (2025). Trends in Daily Nicotine Vaping and Unsuccessful Quit Attempts in Youths. JAMA Network Open, 8(11), e2541061. 10.1001/jamanetworkopen.2025.41061 [PubMed: 41182763]

McConnell R, Barrington-Trimis JL, Wang K, Urman R, Hong H, Unger J, Samet J, Leventhal A, & Berhane K (2017). Electronic Cigarette Use and Respiratory Symptoms in Adolescents. American Journal of Respiratory and Critical Care Medicine, 195(8), 1043–1049. 10.1164/rccm.201604-0804OC [PubMed: 27806211]

Miech RA, Patrick ME, O'Malley PM, Jager JO and Jang JB (2026). Monitoring the Future national survey results on drug use, 1975–2025: Overview and detailed results for secondary school students. (Monitoring the Future Monograph Series). Ann Arbor, MI: Institute for Social Research, University of Michigan. https://monitoringthefuture.org/results/annual-reports/

Mukerjee R, Hirschtick JL, Arciniega LZ, Xie Y, Barnes GD, Arenberg DA, Levy DT, Meza R, Fleischer NL, & Cook SF (2024). ENDS, Cigarettes, and Respiratory Illness: Longitudinal Associations Among U.S. Youth. American Journal of Preventive Medicine, 66(5), 789–796. 10.1016/j.amepre.2023.12.005 [PubMed: 38081374]

National Institutes of Health, National Institute on Drug Abuse. (2023). Reported drug use among adolescents continued to hold below pre-pandemic levels in 2023. Retrieved from https://www.nih.gov/news-events/news-releases/reported-drug-use-among-adolescents-continued-hold-below-pre-pandemic-levels-2023. Accessed on November 1, 2024

Nguyen N, Islam S, Llanes KD, Koester KA, & Ling PM (2024). Classification of patterns of tobacco and cannabis co-use based on temporal proximity: A qualitative study among young adults. Addictive Behaviors, 152, 107971. 10.1016/j.addbeh.2024.107971 [PubMed: 38281461]

Niaura R, Rich I, Johnson AL, Villanti AC, Romberg AR, Hair EC, Vallone DM, & Abrams DB (2020). Young Adult Tobacco and E-cigarette Use Transitions: Examining Stability Using Multistate Modeling. Nicotine & Tobacco Research, 22(5), 647–654. 10.1093/ntr/ntz030 [PubMed: 30820566]

Reboussin BA, Wagoner KG, Ross JC, Suerken CK, & Sutfin EL (2021). Tobacco and marijuana co-use in a cohort of young adults: Patterns, correlates and reasons for co-use. Drug and Alcohol Dependence, 227, 109000. 10.1016/j.drugalcdep.2021.109000 [PubMed: 34507062]

Rose JJ, Krishnan-Sarin S, Exil VJ, Hamburg NM, Fetterman JL, Ichinose F, Perez-Pinzon MA, Rezk-Hanna M, Williamson E, & null, null. (2023). Cardiopulmonary Impact of Electronic Cigarettes and Vaping Products: A Scientific Statement From the American Heart Association. Circulation, 148(8), 703–728. 10.1161/CIR.0000000000001160 [PubMed: 37458106]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Soule E, Bansal-Travers M, Grana R, McIntosh S, Price S, Unger JB, & Walton K (2023). Electronic cigarette use intensity measurement challenges and regulatory implications. Tobacco Control, 32(1), 124–129. 10.1136/tobaccocontrol-2021-056483 [PubMed: 34059553]

Stanton CA, Sharma E, Seaman EL, Kasza K, Edwards KC, Halenar MJ, Taylor KA, Day H, Anic G, Hull LC, Bansal-Travers M, Limpert J, Gardner LD, Hammad HT, Borek N, Kimmel HL, Compton WM, & Hyland A (2020). Initiation of any tobacco and five tobacco products across 3 years among youth, young adults and adults in the USA: Findings from the PATH Study Waves 1–3 (2013–2016). Tobacco Control, 29(Suppl 3), s178–s190. 10.1136/tobaccocontrol-2019-055573 [PubMed: 32321852]

Stevens ER, Xu S, Niaura R, Cleland CM, Sherman SE, Mai A, Karey E, & Jiang N (2022). Youth E-Cigarette Use and Functionally Important Respiratory Symptoms: The Population Assessment of Tobacco and Health (PATH) Study Waves 3 and 4. International Journal of Environmental Research and Public Health, 19(22), 15324. 10.3390/ijerph192215324 [PubMed: 36430043]

Tackett AP, Urman R, Barrington-Trimis J, Liu F, Hong H, Pentz MA, Islam TS, Eckel SP, Rebuli M, Leventhal A, Samet JM, Berhane K, & McConnell R (2023). Prospective study of e-cigarette use and respiratory symptoms in adolescents and young adults. Thorax, thoraxjnl-2022–218670. 10.1136/thorax-2022-218670

Villanti AC, Pearson JL, Glasser AM, Johnson AL, Collins LK, Niaura RS, & Abrams DB (2017). Frequency of Youth E-Cigarette and Tobacco Use Patterns in the United States: Measurement Precision Is Critical to Inform Public Health. Nicotine & Tobacco Research: Official Journal of the Society for Research on Nicotine and Tobacco, 19(11), 1345–1350. 10.1093/ntr/ntw388 [PubMed: 28013271]

Wang JB, Olgin JE, Nah G, Vittinghoff E, Cataldo JK, Pletcher MJ, & Marcus GM (2018). Cigarette and e-cigarette dual use and risk of cardiopulmonary symptoms in the Health eHeart Study. PloS One, 13(7), e0198681. 10.1371/journal.pone.0198681 [PubMed: 30044773]

Weinberger AH, Zhu J, Lee J, Xu S, & Goodwin RD (2020). Cannabis Use and the Onset of Cigarette and E-cigarette Use: A Prospective, Longitudinal Study Among Youth in the United States. Nicotine & Tobacco Research, 23(3), 609–613. 10.1093/ntr/ntaa158

Winhusen T, Theobald J, Kaelber DC, & Lewis D (2019). Regular cannabis use, with and without tobacco co-use, is associated with respiratory disease. Drug and Alcohol Dependence, 204, 107557. 10.1016/j.drugalcdep.2019.107557 [PubMed: 31557578]

Xie C, Jeffers AM, & Winickoff JP (2024). Categorizing Vaping Intensity Among Youth. Nicotine and Tobacco Research, 26(8), 984–990. 10.1093/ntr/ntae003 [PubMed: 38227762]

Xie W, Tackett AP, Berlowitz JB, Harlow AF, Kathuria H, Galiatsatos P, Fetterman JL, Cho J, Blaha MJ, Hamburg NM, Robertson RM, DeFilippis AP, Hall ME, Bhatnagar A, Benjamin EJ, & Stokes AC (2022). Association of Electronic Cigarette Use with Respiratory Symptom Development among U.S. Young Adults. American Journal of Respiratory and Critical Care Medicine, 205(11), 1320–1329. 10.1164/rccm.202107-1718OC [PubMed: 35089853]

Yang H, Li R, Zucker RA, & Buu A (2016). Two-Stage Model for Time Varying Effects of Zero-Inflated Count Longitudinal Covariates with Applications in Health Behaviour Research. Journal of the Royal Statistical Society Series C: Applied Statistics, 65(3), 431–444. 10.1111/rssc.12123 [PubMed: 27041773]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Public Health Significance Statement

This study finds that adolescents and young adults who continue to use e-cigarettes over time exhibit dynamic use patterns, therefore experiencing varying severities of associated respiratory symptoms. This suggests that tobacco control policy should emphasize changing use patterns as well as other tobacco products and cannabis when targeting respiratory symptoms among youth populations.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

E-cigarette Use Categorization at Baseline (Time 1; T1) and 12-month follow up (Time 2; T2) *How recently did you last use an e-cigarette?*			
Never Use	**Ever Use**	**Past 3 or 4-month Use**	**Past 30-Day Use**
Reported no e-cigarette use at both T1 and T2.	Reported use either sometime in the past 6 months, more than 6 months ago, or more than a year ago	Reported use in the past 3 months at T1 or sometime in the past 4 months at T2	Reported at least one day of e-cigarette use in the past 30-days.

Cross- Sequential Use Categories Based on Tl and T2 E-cigarette Categorization	
Never use	Never Used an E-cigarette at both T1 and T2
Non-Escalating	Did not escalate use from Tl to T2 • Past 4 months at T1 to ever used at T2 • Past 30-days at T1 to past 4 months at T2 • Ever at both Tl and T2 • Never-use at T1 to ever use at T2
Increasing	Escalated use from Tl to T2 • Never at T1 to past 4 months use at T2 • Ever at T1 to past 4 months use at T2 • Past 4 months at T1 to past 30-days at T2
Sustained use	Sustained past 30-day use at both Tl and T2

Figure 1.
Categorization of E-cigarette Use.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 1

Sample Characteristics of the Emerging Adulthood Health Study Cohort Stratified by E-cigarette Use Categories

M [SD] or N (%)	Analytic Sample (N=751)	Never Use[a] (n= 371)	Non-Escalating Use[b] (n=120)	Increasing Use[c] (n= 49)	Sustained Use[d] (n= 211)
Gender N (%)					
Male	314 (42.0)	156 (42.3)	50 (41.7)	18 (36.7)	118 (56.2)
Female	423 (56.6)	206 (55.8)	69 (57.5)	30 (61.2)	90 (42.9)
Another Gender Response	11 (1.5)	7 (1.9)	1 (0.80)	1 (2.0)	2 (1.0)
Race N (%)					
NH White	563 (75.3)	257 (69.8)	96 (80.0)	36 (73.5)	174 (82.5)
NH Black	20 (2.7)	13 (3.5)	2 (1.7)	1 (2.0)	4 (1.9)
Hispanic	36 (4.8)	16 (4.3)	9 (7.5)	3 (6.1)	8 (3.8)
NH Asian	94 (12.6)	67 (18.2)	8 (6.7)	6 (12.2)	13 (6.2)
NH Multi-Racial	35 (4.7)	15 (4.1)	5 (4.2)	3 (6.1)	12 (5.7)
Education N (%)		q			
Still in High School	79 (10.5)	64 (17.3)	4 (3.3)	7 (14.3)	4 (1.9)
Not in school, no college degree	24 (3.2)	12 (3.2)	3 (3.3)	0 (0.0)	8 (3.8)
In College	510 (68.0)	221 (59.7)	92 (76.6)	29 (59.2)	168 (79.6)
Advanced degree/in progress	137 (18.3)	73 (19.7)	20 (16.7)	13 (26.5)	31 (14.7)
Age M [SD]	20.56 (2.1)	20.29 (2.3)	20.77 (1.7)	20.83 (2.5)	20.87 (1.7)
Lifetime Asthma Diagnosis N (%)	151 (20.1)	76 (20.5)	26 (21.7)	14 (28.6)	35 (16.6)
Self-Reported COVID-19 (%)	148 (19.7)	53 (14.3)	33 (27.5)	4 (8.2)	58 (27.5)

	Analytic Sample (N=751)		Never Use[a] (n= 371)		Non-Escalating Use[b] (n=120)		Increasing Use[c] (n= 49)		Sustained Use[d] (n= 211)	
	T1	T2	T1	T2	T1	T2	T1	T2	T1	T2
Past 30-Day Tobacco Product Use[e] N (%)	69 (9.2)	58 (7.7)	0 (0.0)	0 (0.0)	16 (13.3)	4 (3.3)	2 (4.1)	5 (10.2)	51 (24.2)	49 (23.2)
Past 30-Day Cannabis Use[f] N (%)	165 (22.0)	180 (24.0)	0 (0.0)	14 (3.8)	55 (45.8)	46 (38.3)	1 (2.0)	5 (10.2)	109 (51.7)	115 (54.5)
Inhalable Cannabis[l] N (%)	160 (97.0)	154 (85.6)	(0.0)	4 (28.6)	53 (96.4)	40 (87.0)	1 (100.0)	3 (60.0)	106 (97.2)	107 (93.0)
Edible Cannabis[m] N (%)	113 (68.5)	88 (48.9)	0 (0.0)	11 (78.6)	45 (81.8)	22 (47.8)	1 (100.0)	3 (60.0)	67 (61.5)	52 (45.2)
Another Way of Using Cannabis[n] N (%)	1 (100.0)	1 (100.0)	0 (0.0)	(0.0)	0 (0.0)	(0.0)	(0.0)	(0.0)	1 (100.0)	1 (100.0)
Number of Days with Symptoms[g] M [SD]	2.21 (4.3)	2.1 (4.4)	1.6 (3.5)	1.6 (3.3)	2.6 (4.6)	2.47 (5.5)	2.0 (4.4)	1.63 (2.8)	3.18 (5.1)	2.81 (5.4)

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Summed Symptoms Average[h,i] M [SD]	0.89 (1.3)	0.79 (1.3)	0.70 (1.2)	0.71 (1.2)	1.1 (1.5)	0.75 (1.3)	0.63 (1.2)	0.80 (1.3)	1.2 (1.4)	0.97 (1.4)
ATSQ Score[j,k] M [SD]	12.0 (4.0)	11.6 (4.2)	11.8 (3.8)	11.2 (3.6)	12.4 (4.8)	11.5 (4.6)	11.6 (3.5)	10.7 (4.0)	12.3 (4.1)	12.3 (4.7)

Note.

a = never used any tobacco product at T1 or T2;

b = did not escalate use from T1 to T2;

c = escalated use from T1 to T2;

d = was a past 30-day user at both T1 and T2;

e = any use of cigarettes, cigars, cigarillos, hookah, pipe tobacco, and smokeless tobacco in the past 30 days;

f = any use of cannabis in the past 30 days;

g = the average number of days experiencing at least one respiratory symptom in the past 30 days;

h = number of participants who experienced said symptom anytime in the past 3 months;

i = on a scale from 0–6;

j = number of participants who experienced said symptom at least once per week;

k = on a scale from 8–40;

l = smoke (joint, blunt, bong, bowl), vape (THC juice, dry herb) or wax;

m = edibles or juice;

n = other cannabis modality.

NH=non-Hispanic.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 2

Respiratory Symptom Frequencies of the Emerging Adulthood Health Study Cohort Stratified by E-cigarette Use Categories

N (%)	Analytic Sample (N=751)		Never Use[a] (n= 371)		Non-Escalating Use[b] (n=120)		Increasing Use[c] (n= 49)		Sustained Use[d] (n= 211)	
	T1	T2	T1	T2	T1	T2	T1	T2	T1	T2
Summed Symptoms[h,i]										
Cough	203 (27.0)	193 (25.7)	73 (19.7)	83 (22.4)	40 (33.3)	30 (25.0)	9 (18.4)	14 (28.6)	81 (38.4)	66 (31.3)
Chest Tightness	66 (8.8)	63 (8.4)	27 (7.3)	26 (7.0)	11 (9.2)	8 (6.7)	4 (8.2)	5 (10.2)	24 (11.4)	24 (11.4)
Chest Pain	63 (8.4)	46 (6.1)	18 (4.9)	18 (4.9)	16 (13.3)	9 (7.5)	2 (4.1)	2 (4.1)	27 (12.8)	17 (8.1)
Shortness of Breath	109 (14.5)	99 (13.2)	43 (11.6)	42 (11.3)	21 (17.5)	14 (11.7)	5 (10.2)	5 (10.2)	40 (19.0)	38 (18.0)
Sore and Dry Throat or Mouth	182 (24.2)	157 (20.9)	77 (20.8)	68 (18.3)	33 (27.5)	24 (20.0)	9 (18.4)	11 (22.4)	63 (29.9)	54 (25.6)
Breathing Difficulties	47 (6.3)	47 (6.3)	24 (6.5)	27 (7.3)	10 (8.3)	6 (5.0)	2 (4.3)	3 (6.1)	11 (5.2)	11 (5.2)
ATSQ Symptoms[j,k]										
Cough first thing in the morning	51 (6.8)	49 (6.5)	21 (5.7)	13 (3.5)	9 (7.5)	8 (6.7)	4 (8.2)	2 (4.1)	17 (8.1)	26 (12.3)
Cough frequently throughout the day	55 (7.3)	52 (6.9)	23 (6.2)	13 (3.5)	9 (7.5)	11 (9.2)	5 (10.2)	3 (6.1)	18 (8.5)	25 (11.8)
Wheezing	32 (4.3)	21 (2.8)	14 (3.8)	7 (1.9)	10 (8.3)	6 (5.0)	1 (2.0)	0 (0.0)	7 (3.3)	8 (3.8)
Shortness of breath when walking	79 (10.5)	98 (13.0)	37 (10.0)	39 (10.5)	15 (12.5)	17 (14.2)	4 (8.2)	4 (8.2)	23 (10.0)	38 (18.0)
Shortness of breath during exercise or walking upstairs	254 (33.8)	208 (27.7)	112 (30.2)	101 (27.2)	42 (35.0)	28 (23.3)	16 (32.7)	12 (24.5)	84 (39.8)	67 (31.8)
Phlegm or mucous when you cough	60 (8.0)	58 (7.7)	29 (7.8)	21 (5.7)	9 (7.5)	12 (10.0)	5 (10.2)	4 (8.2)	17 (8.1)	21 (10.0)
Pain or tightness in the chest	54 (7.2)	38 (5.1)	26 (7.0)	16 (4.3)	9 (7.5)	6 (5.0)	3 (6.1)	2 (4.1)	16 (7.6)	14 (6.6)
Getting very tired in a short time	154 (20.5)	133 (17.7)	67 (18.1)	54 (14.6)	29 (24.2)	23 (19.2)	9 (18.4)	7 (14.3)	49 (23.2)	49 (23.2)

Note. T1 = baseline. T2 = 12-month follow-up.

a = never used any tobacco product at T1 or T2.

b = did not escalate use from T1 to T2.

c = escalated use T1 to T2.

d = used e-cigarettes in the past 30 days at both T1 and T2.

h = number of participants who experienced said symptom anytime in the past 3 months.

i = on a scale from 0–6.

j = number of participants who experienced symptom at least once per week.

k = on a scale from 8–40.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 3.

Unadjusted and Adjusted Binary Logistic Regression Models of E-cigarette Use Categories, Respiratory Symptoms, and ATSQ Scores.

User Category	OR	95% CI	p-value	aOR[+]	95% CI	p-value
Summed Symptoms (Any Symptoms, No Symptoms)						
Never use	Ref	—	—	Ref	—	—
Sustained use	1.47	[1.04, 2.07]	0.03[*]	1.18	[0.76, 1.82]	0.46
Non-Escalating use	1.07	[0.70, 1.64]	0.74	0.86	[0.53, 1.39]	0.54
Increasing use	1.13	[0.61, 2.08]	0.69	1.08	[0.58, 2.00]	0.80
Days with Symptoms in the Past 30 Days (Yes, No)						
Never use	Ref	—	—	Ref	—	—
Sustained use	1.64	[1.16, 2.32]	0.01[*]	1.22	[0.92, 1.97]	0.37
Non-Escalating use	1.06	[0.68, 1.63]	0.80	0.79	[0.48, 1.30]	0.37
Increasing use	1.23	[0.65, 2.27]	0.51	1.18	[0.62, 2.19]	0.61
ATSQ (Any Symptoms, No Symptoms)						
Never use	Ref	—	—	Ref	—	—
Sustained use	1.12	[0.74, 1.70]	0.59	0.86	[0.52, 1.42]	0.54
Non-Escalating use	0.92	[0.56, 1.53]	0.62	0.74	[0.43, 1.31]	0.30
Increasing use	0.83	[0.40, 1.79]	0.40	0.80	[0.39, 1.74]	0.56

Note. N = 751.

[*] = $p < .05$.

Ref = Reference category.

[+] = Adjusted analyses controlled for any use of cigarettes, cigars, cigarillos, hookah, pipe tobacco, and smokeless tobacco in the past 30 days, any use of cannabis in the past 30 days, lifetime COVID-19, and lifetime asthma diagnosis.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

J Stud Alcohol Drugs. Author manuscript; available in PMC 2026 April 20.

Table 4

Adjusted Zero-Inflated Poisson and Linear Regression Models for Summed Symptoms, Average Symptoms in the Past 30 Days, and ATSQ Respiratory Outcomes.

User Category	Estimate	SE	p-value
Zero-inflated Model Coefficients of Summed Respiratory Symptoms (N = 751)			
Never use[a]	Ref	—	—
Sustained use[a]	−0.11	0.28	0.69
Non-Escalating use[a]	0.21	0.32	0.52
Increasing use[a]	−0.08	0.42	0.85
Poisson Model Coefficients of Summed Respiratory Symptoms (N = 751)			
Never use[b]	Ref	—	—
Sustained use[b]	0.13	0.14	0.37
Non-Escalating use[b]	0.03	0.18	0.86
Increasing use[b]	0.02	0.22	0.93
Linear Regression Coefficients of Days with Symptoms in the Past 30 Days; N = 269, $F(6,262) = 2.56$, $p = 0.02$, Adjusted $R^2 = 0.03$			
Never use[c,d]	Ref	—	—
Sustained use[c,d]	2.05	0.95	0.03*
Non-Escalating use[c,d]	2.73	1.17	0.02*
Increasing use[c,d]	−0.72	1.40	0.61
ATSQ Scores			
Linear Regression Coefficients of ATSQ Scores; N = 420, $F(5,414) = 6.15$, $p < .001$, Adjusted $R^2 = 0.058$			
Never use[c,e]	Ref	—	—
Sustained use[c,e]	1.48	0.54	0.0006*
Non-Escalating use[c,e]	0.73	0.61	0.24
Increasing use[c,e]	−0.45	0.83	0.59

Note. Adjusted analyses for summed symptoms and frequency of symptoms in the past 30 days controlled for any use of cigarettes, cigars, cigarillos, hookah, pipe tobacco, and smokeless tobacco in the past 30 days, any use of cannabis in the past 30 days, lifetime COVID-19, and lifetime asthma diagnosis. Analyses of ATSQ scores excluded participants with asthma and controlled only for tobacco and cannabis use in the past 30 days.

[a] = Zero-inflated Model Binomial Coefficients.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

b = Poisson Model Coefficients.

c = Linear Regression Models.

d = Model only includes participants that indicated they experienced days with respiratory symptoms.

e = Model only includes participants that had non-zero ATSQ scores and excluded participants with asthma.

* = $p < .05$.

Ref = Reference category